7/5/25 (E)

SOMAIYA VIDYAVIHAR UNIVERSITY

Semester: January 2025 –April 2025		
Maximum Marks: 100 Examination: ESE Examination		Duration:3 Hrs.
Programme code: 01 Programme: BTech Computer Engineering	Class: TY	Semester: VI(SVU 2020)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: COMP	
Course Code: 116U01E628 Name of the Course: Applied Cryptography		
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	20
i)	Define confusion and diffusion. Comment on their significance in cryptography.	5
ii)	Explain any two major attacks on RSA.	5
iii)	Define the following concepts and comment in one line about their significance in data security: a. Initialization vector b. P-boxes c. Fiestel cipher d. Avalanche effect e. Cryptanalysis	5
iv)	Explain the following concepts with respect to the applications of prime number theory in cryptography; give examples to support your answer. a. Euler's totient function b. Euclidean Algorithm	5
v)	Discuss the requirements of cryptographic hash functions.	5
vi)	Explain the Key generation process in DES.	5
Q2 A	Solve the following	10
i)	Encrypt the message "MEET ME IN HALL" using railfence cipher of height =3.	5
ii)	What are the major weaknesses of DES? Is there any possible way to overcome them? Discuss the solutions, if there are any.	5
	OR	
Q2 A	A university uses a basic password-based login system to protect access to its student records portal. Each student has a unique ID and password to access personal academic information, such as grades, attendance, and course registration. The system does not enforce strong password policies (e.g., no minimum length or complexity requirements), and there is no multi-factor authentication. Additionally, the system does not lock users out after repeated failed login attempts. Using the Motive, Opportunity, and Method (MOM) model, analyze how this system can be exploited by various users and using various methods.	10
Q 2 B	Solve any **One**	10
i)	A. Compute: a. Additive inverse of -9 mod 17 b. Multiplicative inverse of 23 modulo 120	10

	c. Multiplicative inverse of 20 modulo 120 B. Explain one round of DES with a suitable diagram.	
ii)	Compare and contrast – Symmetric key cryptography and asymmetric key cryptography. Which cryptosystems will be followed in case of: 1. Encryption of session keys 2. Encryption of logical hard drive/folder 3. Encryption of password 4. Encryption of classified data folders	
Q3	Solve any **Two**	**20**
i)	Explain Deffie Hellman key exchange method. **State** limitation of the DH scheme. How it can be improved with Quantum Key Distribution protocol?	6+2+2
ii)	In a RSA symmetric encryption scenario, assume n=33. Following the RSA constraints, a. Choose the public key e. b. Compute the private key d. c. Encrypt M=2 and compute ciphertext for the same.	10
iii)	Explain Kerberos as authentication protocol. State two real life applications wherein its in use.	4+3+3
Q4	Solve any **Two**	**20**
i)	Decode the following for the security goal(s) achieved: a. Sender to receiver: Message + $E_{Kpriv-sender}$[Hash(Message)] + Timestamp b. Sender to receiver: $E_{K-Shared}$[Message] + $E_{Kpublic-receiver}$[Hash(Message)] + Random Number Receiver to sender: Square[mid three digits of received random number] mod P(pre-decided number) c. Sender to receiver: $E_{K-Shared}$[Message] + $E_{Kpublic-Receiver}$ [$E_{Kpriv-sender}$[Hash(Message)]] + Timestamp+OTP Also consider- replay attacks, man in middle attacks, repudiations on sender and receiver side,etc.	10
ii)	Discuss working of HMAC for Message authentication and integrity.	3+3+4
iii)	Consider a problem instance as: A network of banks wants to detect fraud patterns in their combined transaction data without revealing customer details to one another. They run a joint analysis where each bank inputs its data, but no individual bank sees the others' data — only the final results are shared. To log in to the system securely, analysts must prove their identity without entering any password, and the proof process does not reveal any sensitive login details. Additionally, one bank is testing a communication system that uses the properties of quantum physics to ensure that encryption keys cannot be intercepted without detection. **Question**: Identify the advanced cryptographic methods those could be used here. Explain how each method contributes to the privacy or security of the system.	10
Q5	Solve any **four**	**20**
i)	Write a short note on- Vulnerabilities, zero day exploits and controls.	5
ii)	Write short note on- Applications of Stream ciphers and block ciphers	5
iii)	Discuss working of Digital Authentication Algorithm[DAA] and its applications.	5
iv)	Discuss X.509 Digital certificates and their significance in authentication.	5
v)	MD5: applications, strengths and weaknesses.	5
vi)	Cryptanalysis of substitution ciphers.	5